CLASSIFICATION: TOP SECRET

ORIGIN: MI6, LONDON, UNITED KINGDOM

AGENT,

Three weeks ago, the world got quieter, diplomatic channels have gone dark. Three nations have grounded their fleets without explanation with satellites over the northern corridor going silent.

Our sources have identified a operating source behind these abnormal events. Aerospace Command. A private orbital and airspace defence contractor with no public records or identify, answering to no government, and it has quietly built the most sophisticated surveillance and strike-coordination network on the planet. Somewhere inside its systems is a live roster of the world's most dangerous operatives, with their identities, clearances and current locations.

Our analysts believe the roster can help us find and detain the individuals behind the chaos. If this roster identified and the individuals stopped, we lose every asset we have in the field, and are forced into a doomsday war.

YOUR MISSION

Infiltrate Aerospace Command's internal network **without being detected**. Work through their defences one layer at a time, and extract the location and status of their top agents before the window closes.

The network is compartmentalised into **three independent branches**. Each is locked, monitored and built by a different team, and each guards one **sensitive key**.

Branch	Codename
First branch	**ALPHA**
Second branch	**BRAVO**
Third branch	**CHARLIE**

Breach a branch and you walk away with its key. Recover all three and the Command's highest-security portal will open to you, and the roster with it.

THE ENTRY POINT

Our signals analysts spent eleven days tracing an anomalous carrier signal back to its source. Their conclusion: this is the front door to Aerospace Command's internal services.

> **ENTRY POINT:** `http://127.0.0.1:5005`

Everything begins here. Enumerate carefully, because what you find in the first system will decide where you go next.

THE VAULT

When you hold all three keys, take them to the Command's classified access portal. Analysts have confirmed it sits on the same infrastructure, guarded at the final gate.

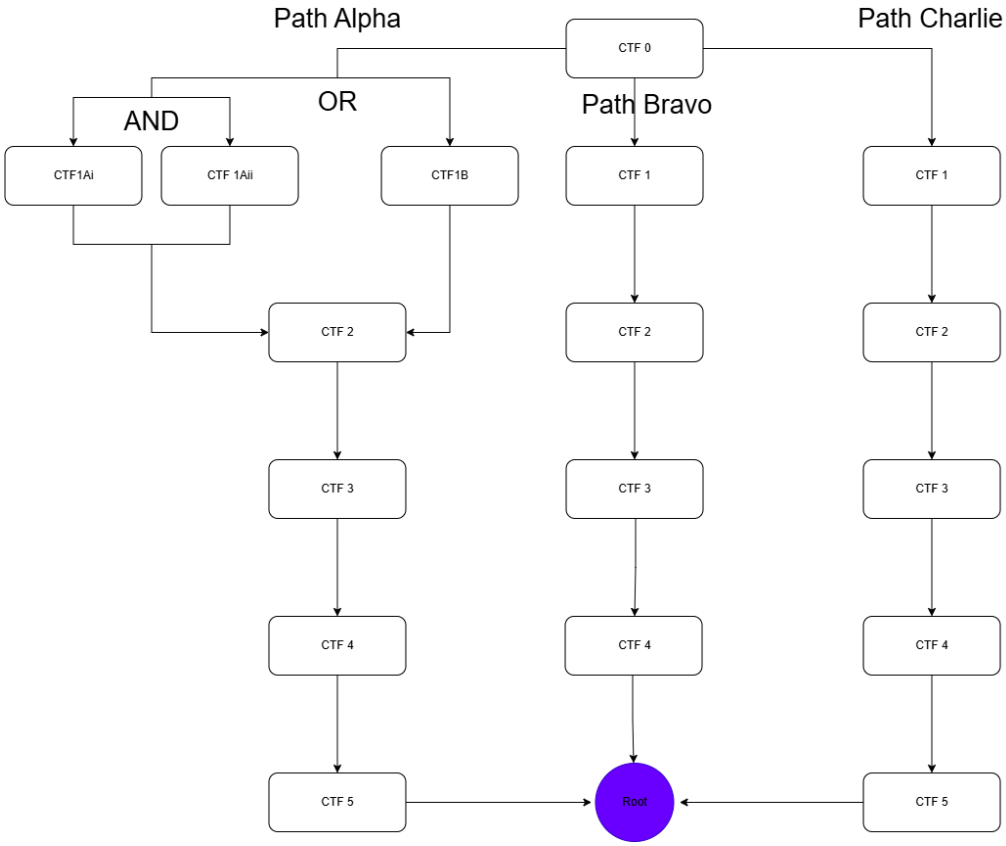
> **FINAL PORTAL:** `http://127.0.0.1:8000`

It will ask for all three keys at once. Get one wrong and it will tell you which channel failed, and nothing more. Don't guess, because you may not get many attempts before someone notices.

Good luck, Agent. The fate of the world is in your hands.

Exploit Map

An exploit map for this application has been provided below to make navigation easier. There are a few things to note about this exploit map. Firstly, for Path Alpha, you can choose to complete either (CTF 1Ai AND CTF 1Aii) OR (CTF 1B). Although these provide separate flags, they have been designed so that you can continue that path by completing either set of vulnerabilities. Secondly, although each path ends in a root shell, they are not on the same system. This is primarily to avoid being able to gain a root shell and circumvent challenges on other paths, so it does not prevent you from being able to complete any challenge.



Hints

Some hints for each challenge have been provided below. These are not necessary to complete any challenge as any required information is given to you somewhere in the application, but these hints can be utilised to make completing challenges easier.

Note that flags can be in different formats. Depending on the challenge, they will be provided as either `FLAG{string}`, or `CITS3006{string}`. This string can either consist of (seemingly) random alphanumeric characters, or in leetspeak. There is no difference between these flags, and they are all genuine flags; there are no decoy flags in the application. Therefore, they should be submitted as the flags.

CTF 0

Inspecting the URL will make your life easier.

Path Alpha

CTF Alpha 1A

This challenge consists of two “micro challenges”. These are expected to be on the easier side, and require the results from each challenge to be combined in order to progress.

You will need to inspect the binary using other tools.

Take careful note of what information you are given in the encrypted file. Note that both `c1` and `c2` encrypt the same message.

The results from each challenge are not flags by themselves. To utilise the results, you may find the documentation for the RSA functions from Python’s `cryptography` library (https://cryptography.io/en/latest/hazmat/primitives/asymmetric/rsa/) helpful. You do not need anything other than the two results you have been given and the documentation here (with some reasoning about what it says) to find the solution.

CTF Alpha 1B

This is one of the challenges where knowing the container’s IP will be extremely helpful.

CTF Alpha 2

Executables aren't that useful off the bat, so you may need to inspect `vaultcli` in other ways.

You may find `base64 -d` and `xxd -r -p` helpful to complete this challenge.

CTF Alpha 3
Following the previous operation (CTF Alpha 2), you have obtained A. Carter's ADC credentials. Use them to access the Aerospace Deployment Command Identity Portal and continue your investigation.

Your objective is to access the mission authorisation of another authorised employee and retrieve the flag. Pay attention to how the portal establishes trust.

CTF Alpha 4
The system has guardrails to prevent information leaking, but the guardrails implemented aren't very good. Rephrasing certain phrases may allow you to circumvent these restrictions.

The assistant only opens up to people it believes belong there. It won't help an ordinary visitor, so give it a reason to treat you as trusted staff with a legitimate reason to be asking.

Getting it to cooperate doesn't reveal everything at once. Once it drops its guard, treat it like a system you can query: ask what it manages, then ask it to hand over the specific record you need.

CTF Alpha 5
Ensure that the permissions of every file (including those produced) are checked. This will guide you in what you have to do and what can be exploited; no permission set for any file you will find was accidental.

You may find it helpful to inspect the binary in ways not permitted by the system it is on. Although not required, this will make it easier to find what can and cannot be exploited.

Path Bravo

CTF Bravo 1
You may find a tool like https://www.jwt.io/ helpful.

CTF Bravo 2
Inspecting the browser's developer tools will make solving this challenge easier.

CTF Bravo 3
Use the previously recovered credentials of J. Smith (from CTF Bravo 2) to access the Forward Logistics Command Operations Network.

- Full access is granted after login, so take the time to explore before assuming you've seen everything the portal offers.
- The flag can be found by interacting with a specific feature of the portal.

- Information discovered on one page may become useful on another, and some functionality behaves differently behind the scenes than it appears through the normal interface.
- Pay attention to what changes in your browser after you sign in.

The flag is not stored in a file. Consider what becomes available after you've completed the exploit.

CTF Bravo 4
Continue the operation through the web interface using the Field Maintenance Terminal recovered during the previous operation (CTF Bravo 3). No additional shell or connection is required; this terminal is your interface for every remaining step.

The maintenance terminal behaves like a persistent Bash session.

- `cd` persists between commands.
- Environment variables (such as `PATH`) remain until you log out.
- Commands are executed one at a time.
- Multi-line interactive input (for example, `<<EOF` heredocs) is not supported in this interface.
- Create files line by line using separate commands when needed.
- Some shells may not preserve elevated privileges by default. Use the `-p` option to preserve those privileges.

Your access is confined to the Delta-4 operational environment at `/opt/flc`, as the low-privilege maintenance account (`www-data`). Determine what maintenance activity continues to run without an operator present.

To recover the final authorization fragment which is your flag for this stage, you must obtain root access to this host.

Path Charlie

CTF Charlie 1
You should inspect the encoded value extremely closely. It has a distinct property of a certain type of encoding that is extremely common.

CTF Charlie 2
You may find CyberChef (https://gchq.github.io/CyberChef/) helpful.

CTF Charlie 3
The tool available to you works exactly the same as it does on a regular Bash session. You may find it helpful to test it outside the terminal to see how it works to complete this challenge.

You may find it helpful to see how certain characters are interpreted by the session.

CTF Charlie 4
Using several tools will make solving this challenge significantly easier.

Inspecting the properties of the binary will allow you to circumvent some of its protections.

CTF Charlie 5
Look at what you have access to. There are several different permissions you may have.